

Job Processing

Tags:

Time Limit: 1 s Memory Limit: 128 MB
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Codes

AI Code Tips

Description

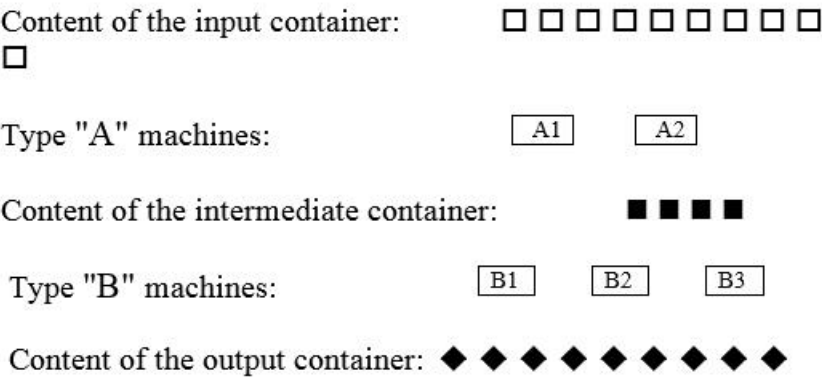


Figure 1

A factory is running a production line. Two operations have to be performed on each job: first operation "A", then operation "B". There is a certain number of machines capable of performing each operation. Figure 1 shows the organisation of the production line that works as follows. A type "A" machine takes a job from the input container, performs operation "A" and puts the job into the intermediate container. A type "B" machine takes a job from the intermediate container, performs operation "B" and puts the job into the output container. All machines can work in parallel and independently of each other, and the size of each container is unlimited. The machines have different performance characteristics, a given machine works with a given processing time.

Give the earliest time operation “A” can be completed for all N jobs provided that the jobs are available at time 0. (Subtask A). Also compute the minimal amount of time that is necessary to perform both operations on N jobs (Subtask B).

Input

The input contains positive integers in five lines. The first line contains N, the number of jobs ($1 \leq N \leq 1000$). On the second line, the number M1 of type "A" machines ($1 \leq M1 \leq 30$) is given. In the third line there are M1 integers, the job processing times of each type "A" machine. The fourth and the fifth line contain the number M2 of type "B" machines ($1 \leq M2 \leq 30$) and the job processing times of each type "B" machine, respectively. The job processing time is measured in units of time, which includes the time needed for taking a job from a container before processing and putting it into a container after processing. Each processing time is at least 1 and at most 20.

Output

Your program should write two lines. The first line should contain one positive integer: the solution of subtask A. The second line should contain the solution of subtask B.

Samples

input	Copy
<pre>5 2 1 1 3 3 1 4</pre>	
output	Copy
<pre>3 5</pre>	

Source

[IOI1996 Day1](#)

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[HZNUOJ](#) is based on [HUSTOJ](#)

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Server Time: 2025-8-26 15:41:35